2 Human Body: The Skeletal System

We know that a group of organs working together to perform a particular function is called an organ system. We will learn about another organ system known as the skeletal system in this chapter.

Learn about

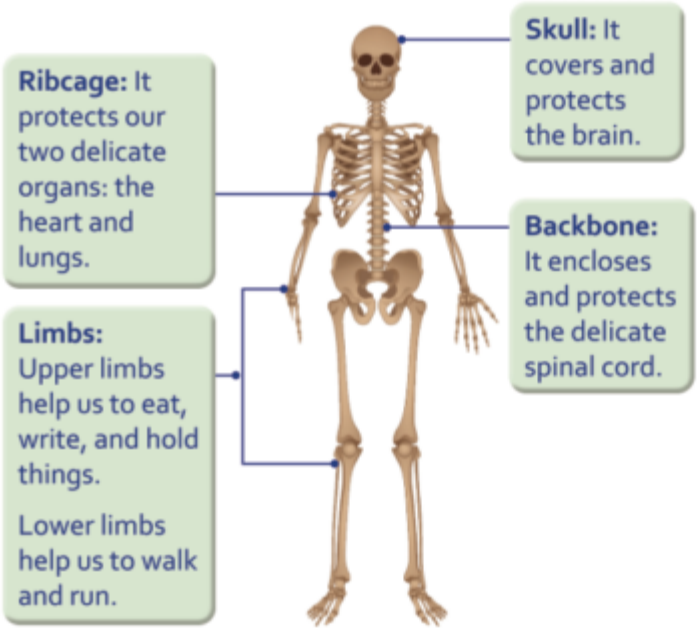
- The skeletal system
- Major bones of the human body
- Functions of bones in the human body
- Joints
- Muscles
- Importance of the skeletal system
- Care of bones and joints

The Skeletal System

The skeletal system consists of the bones present in our body. These bones form a rigid framework that supports the body and gives it a definite shape. It also performs other important functions, such as helping us move, bend, and protecting our delicate internal organs. This rigid framework of bones is called the **skeleton**.

Bones

There are 206 bones in an adult human skeleton. Bones are mainly made up of minerals such as calcium and phosphorus. Calcium makes the bones strong. The bones are thus strong enough to support the entire weight of our body.



Skeletal system

Fact File

The stapes is the smallest bone in the body. It is present in the ear and is 3 mm long. Along with two other bones, incus and malleus, it helps in hearing.

Types of Bones

The human body has different types of bones that have different shapes and sizes.

- Bones of the limbs, including those of the fingers and toes, are long.
- Bones of the wrist and ankle are short.
- Most of the bones of the skull and the breastbone are flat in shape.
- The bones of the spine, hips, and some of the bones of the skull are irregular in shape.

Bones of the limb are long.

The breastbone is flat.

The bone of the wrist is short.

The bone of the spine is irregular in shape.

▶ Major Bones of the Human Body

The skeleton consists of the skull, backbone, ribcage, limbs, jawbone, and girdle bones.

Skull

The skull is a hollow, round structure present in the head.

Structure of Skull

It is made up of 22 bones, of which eight bones form the part of the skull (that protects the brain) and the remaining 14 bones form the face. In addition, there are six small bones in our ears: three in each ear. All the bones of the skull are fixed and immovable, except the lower jaw. The lower jaw is joined to the upper jaw. The upper and lower jaws have teeth.

Skull

Functions of Skull

- The skull covers and protects the brain.
- It also gives shape to the face.
- The movable lower jaw helps us to talk and eat (teeth).

Jawbone

The jawbone is the U-shaped bone of the lower jaw.

Structure of Jawbone

It is the largest and strongest bone in the skull. It is the only bone in the skull that can move.

Function of Jawbone

It helps us to move our mouth so that we can chew food.

Backbone

The backbone is a long and flexible bony structure. It is also known as the **vertebral column** or **spine**. It is connected to the base of the skull at its top.

Backbone

Structure of Backbone

It is made up of 33 small, ring-like bones. The individual small bones are called **vertebra**. The delicate spinal cord passes through them.

Function of Backbone

The backbone encloses and protects the delicate **spinal cord**.

Fact File

Animals with a backbone (vertebral column), such as humans, tigers, cows, lions, and dogs, are called vertebrates. Animals without a backbone and skeletal system, such as earthworms, snails, cockroaches, and butterflies, are called invertebrates.

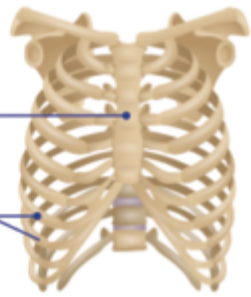
Ribcage

The ribcage is a bony structure present in the chest region.

Structure of Ribcage

It is made up of 12 pairs of thin, curved bones called **ribs**. A long, flat bone, called the **sternum**, is present at the centre of the chest and holds these ribs together in the front. At the back, the ribs are held together by the spine or backbone. However, the last two pairs of ribs are not attached to the sternum.

These two pairs of ribs are called the **floating ribs**.



Ribcage

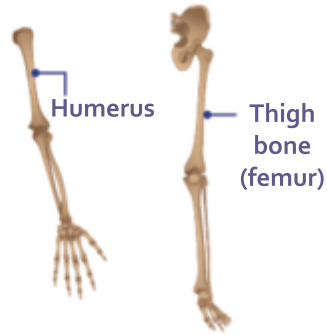
Function of Ribcage

The ribcage protects the delicate organs such as the heart and the lungs.

Limbs

Human beings have four limbs: a pair of upper limbs (arms) and a pair of lower limbs (legs).

Upper limbs Upper limbs have two parts: the upper arm and the lower arm. The bone present in the upper arm, called **humerus**, is joined to the two bones of the lower arm at the elbow region.



Limbs

Function of Upper Limbs

The bones, muscles, and joints in the upper limbs enable movement of the arms, and thus help us to perform activities such as eating, writing, and lifting objects.

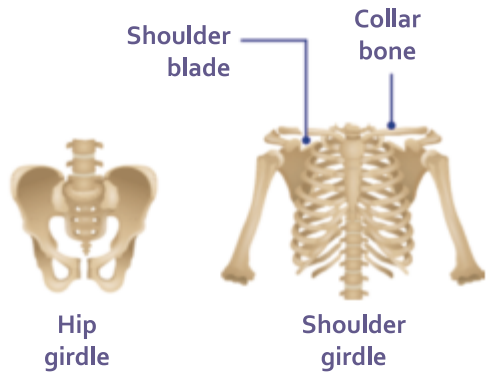
Lower limbs Lower limbs have two parts: the upper leg and the lower leg. The upper leg has the longest bone in the body, called the **femur**. The femur is also known as the **thigh bone**. It is joined to the two bones of the lower legs at the knees.

Function of Lower Limbs

The bones, muscles, and joints of the lower limbs enable movement of the legs, and thus help to walk and run.

Girdles

A girdle is a bony, ring-like structure. There are two girdles present in our body: the shoulder girdle and the hip girdle. The shoulder girdle is made up of the shoulder blade and the collar bone. The hip girdle is made up of three bones fused to form a single bone.



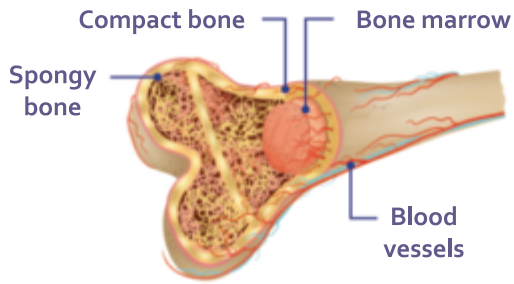
Girdles

▶ Functions of Bones in the Human Body

The main functions of the bones that form the skeleton are listed as follows.

Shape and support It gives shape and support to the body.

Protection It protects the delicate internal organs such as the brain, spinal cord, heart, and lungs.



Bone marrow

Movement It allows the movement of the various parts of the body.

Production and storage Bone marrow present inside the bones produces the red blood cells. The bones of the skeleton also store the mineral calcium.

▶ Joints

*A **joint** is a place in the skeleton where two or more bones connect.* Bones are held together at the joints with the help of **ligament**. At the joints, the surfaces of the bones are covered with cartilage.

Cartilage

***Cartilage** is a strong, elastic, yellowish or white-coloured tissue present around the joints.* Cartilage can bend. Organs such as our nose and ears also have cartilage.

Functions of cartilage Cartilage protects the bones from getting damaged as a result of the constant[1] movements that take place at the joints.

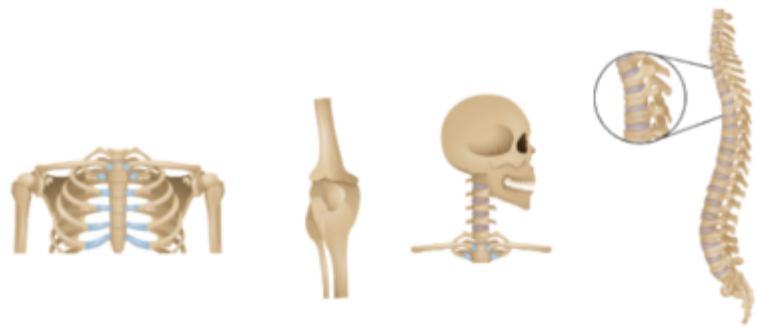
Types of Joints

Joints are of two types: movable and immovable. All the joints except those in the skull are movable joints.

Types of Movable Joints

Ball-and-socket joint In this type of joint, the ball-shaped end of one bone fits into the cup-shaped socket of another bone. This type of joint allows movement in all directions. The shoulder and the hip joints are examples of ball-and-socket joints.

Hinge joint This type of joint allows only back and forth movements. Hinge joints are found in the knees, elbows, fingers, and toes.



Ball-and-socket joint (shoulder) | Hinge joint (knee) | Pivot joint (neck) | Gliding joint (backbone)

Pivot joint This type of joint allows us to move our head sideways. Pivot joints are found between the first and the second vertebrae of the neck region of the backbone.

[1]**constant:** continuous

Gliding joint This type of joint allows the bones to glide[2] against each other. Gliding joints are found in the bones of the wrist and ankle, and between the bones of the vertebral column.

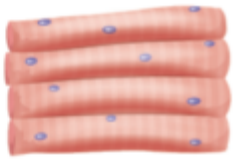
Importance of Joints

- The bones present in the human body are rigid and cannot bend. The joints make our body flexible by allowing movement in different directions.
- There are many different types of joints. Each of these types allows movement in a certain manner.

▶ Muscles

Muscles are the soft tissues present in various places in our body. They are present around bones, joints, and even inside our organs. They do everything from moving our bones and organs to contracting and relaxing our heart. There are over 600 different muscles in our body. They remain attached to the bones through tendons. ***Tendons*** *are thin, chord-like structures that connect bones with muscles.* Based on their function, muscles can be of the following two types.

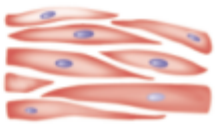
Voluntary muscles Voluntary muscles are those muscles whose movements can be controlled by us. The skeletal muscles that are present in the arms, legs, hands, and feet are examples of voluntary muscles.



Voluntary muscles

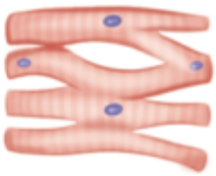
Involuntary muscles Involuntary muscles work on their own and their movements cannot be controlled by us. The following are the two types of involuntary muscles present in the human body.

- **Smooth muscles** *The involuntary muscles present inside the walls of the internal organs, such as the stomach and intestines, are* ***smooth muscles****.* Their contraction and relaxation cause movement within the organs. Food passes through the food pipe and the stomach due to the contraction and relaxation of smooth muscles.



Smooth muscles

- **Cardiac muscles** *The involuntary muscles present inside the walls of the heart are called* ***cardiac muscles****.* These muscles work all the time, pumping blood to different parts of the body, whether we are asleep or awake.



Cardiac muscles

[2]glide: to move smoothly

Questions

Name the following.

1. The place in the skeleton where two or more bones connect .....
2. The elastic, yellowish or white substance present around the bones .....
3. The type of joint that allows movement in all directions .....
4. The thin chord-like structure that connects bone with muscles .....
5. The type of muscles whose movements can be controlled by us .....

Importance of Muscles

- Muscles provide support to the bones.
- They assist the skeleton in moving the limbs and the other parts of the body.
- Muscles inside the internal organs help them in carrying out important functions. For example, the cardiac muscles help the heart in pumping blood to different parts of the body.

▶ Importance of the Skeletal System

The skeletal system includes all the bones, joints, and muscles present in the body. Let us discuss the importance of these parts of the skeletal system.

Importance of Bones

- They provide shape and support to the body.
- They protect the delicate internal organs of the body.
- They allow movement of various parts of the body.
- The bone marrow present inside the bones produces the red blood cells.
- Bones also store the mineral calcium.

▶ Care of Bones and Joints

For our body to be able to move and function properly, we need to take good care of the bones and joints present in it. To make our bones and joints strong and healthy, we should do the following:

Some food items that contain proteins

- We should eat food items that are rich in **proteins**. Proteins help in the growth and repair of the body. They also help in building muscles. Food items that are rich in proteins are also called body-building foods. Young children need

a more protein-rich diet than adults to help in their growth and development. Milk, eggs, cheese, fish, meat, beans, and pulses are good sources of proteins.

- We should eat food items that are rich in **calcium**. It helps in the formation of bones and teeth. Some examples of calcium-rich foods are milk, curd, almond, and cheese.
- Vitamins and minerals protect our body from diseases and help in making bones, joints, and muscles strong and healthy. Some important vitamins that our body needs to maintain strong bones, joints, and muscles are vitamins C and D. Vitamin C helps in repairing the cartilages in joints, while vitamin D helps the body to absorb the calcium present in food. Examples of food that are rich in vitamin C include all citrus fruits, dark green leafy vegetables, and tomatoes. Fatty fish, cheese, and egg yolks are some sources of vitamin D. Fresh fruits and vegetables, milk, eggs, and peanuts are rich sources of vitamins and minerals.

Some food items that contain vitamins and minerals

- We should exercise regularly. Some physical activities that are good for bones and joints include walking, running, and playing sports such as football, basketball, and tennis. Some simple yoga exercises that involve balance and stretching are also good for the bones, joints, and muscles.

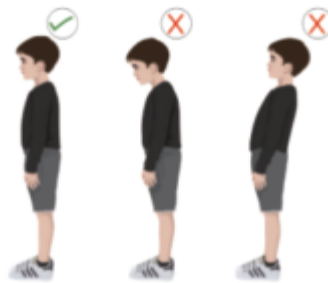
Regular exercise

Importance of Rest and Good Posture

To help our body to function properly, we must make sure that it gets enough **rest**. The body needs 6 to 8 hours of sleep every day. Lack of proper sleep can lead to fatigue and illness. Therefore, we must not stay awake for too long at night.

Correct posture while sitting

Posture is the way in which we keep our body while standing or sitting. It is important to keep our backs straight while sitting or standing. It is not advised to sit or stand with our back bent. A bent back puts burden on the backbone and joints. This may lead to joint and muscle pain later.



Correct posture while standing

Wrap Up

- Bones form a framework called skeleton that gives shape to the body.
- There are 206 bones in an adult human skeleton.
- Bones are made of calcium and phosphorus.
- Bones are held together at the joints with the help of ligament.
- Some major bones of the human body are the skull, backbone, ribcage, limbs, jawbone, and girdle bones.
- The major functions of bones include providing shape, support, and protection, allowing movement, production of red blood cells, and storage of calcium.
- A joint is a place in the skeleton where two or more bones connect.
- There are 600 different muscles in our body.
- Bones, joints, and muscles are important to us because bones provide shape and support, joints allow movement of bones in different directions, and muscles provide support to the bones.
- To make bones, joints, and muscles strong, we need to include food items that are rich in proteins, vitamins, and minerals to our diet.
- Posture is the way in which we keep our body while standing or sitting.
- Regular exercise keeps the bones, joints, and muscles, strong and healthy.

Exercises

SECTION I

REMEMBERING UNDERSTANDING APPLYING

A **Choose the correct option.**

1. The number of bones in an adult human skeleton is .....
   a. 210 b. 208 c. 206 d. 201
2. The bones of the wrist and ankle are .....
   a. long b. short c. tall d. fat
3. Heart and lungs are protected by the .....
   a. ribcage b. skull c. backbone d. girdle
4. The jawbone is the ..... bone of the lower jaw.
   a. A-shaped b. I-shaped c. U-shaped d. Y-shaped
5. Where do we find voluntary muscles?
   a. lungs b. heart c. legs d. stomach

B Assertion and Reasoning questions

1. **Assertion (A):** The skeletal system provides support and structure to the body.

   Reason (R): The skeletal system is made up of long, rigid bones.

   a. Both A and R are True
   b. Both A and R are False
   c. A is True and R is False
   d. A is False and R is True

2. **Assertion (A):** Cartilage helps to protect bones at the joints.

   Reason (R): Cartilage is a hard and inflexible material.

   a. Both A and R are True
   b. Both A and R are False
   c. A is True and R is False
   d. A is False and R is True

C Choose the correct option to fill in the blank.

1. All the bones present in our body form our ..... (digestive/skeletal) system.
2. The bones of the limbs are ..... (short/long).
3. The ..... (ribcage/skull) protects the brain.
4. The upper jaw ..... (or/and) lower jaw have teeth.
5. Bones are held together by ..... (tendons/ligament).

SECTION II

D Short answer questions

1. Name the minerals present in our bones.
2. Define skeleton. Also, mention its role.
3. What is the function of ribcage?
4. What is vertebra? Where is it present?
5. Differentiate between voluntary and involuntary muscles.
6. Write any two points related to the care of bones and joints.

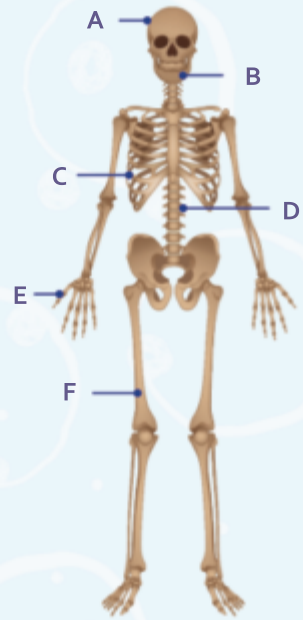
E Long answer questions

1. What are the major functions of bones in the human body?
2. Mention the different types of movable joints present in our body.
3. What is the importance of joints and muscles?
4. What kinds of food act like building materials to keep our bones and muscles strong?
5. Neha's teacher keeps reminding her to sit up straight in the class. Why is good posture important? How does it affect your overall health?

Picture Study

TP REMEMBERING UNDERSTANDING APPLYING

Look at the diagram and answer the questions that follow.



1. What does the given diagram depict?
   It depicts our ..... system.
2. Label the parts A, B, C, D, E, and F in the diagram.
3. What is the shape of part B? Is it movable or immovable?
   .....
4. Name any one bone of part E.
   .....
5. Part F is made up of the longest and strongest bone. Can you name it?
   .....
6. Write the functions of:
   Part A .....
   Part C .....
   Part D .....

My Learning Corner

TP ANALYSING EVALUATING CREATING

A Think about

1. Nitin asked his father, "Papa, why does our skeleton need to be hard?" Complete the conversation.
   His father answered: .....
   .....
2. Reema was wondering why we need joints and muscles. Can you explain it to her?
   .....
   .....

B Try out

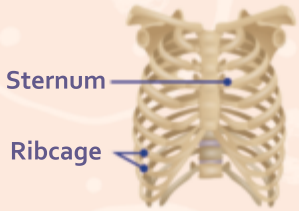
1. Create a blog post highlighting the importance of bone health. Use creative visuals and infographics to showcase healthy habits for strong bones and joints.

TECH INTEGRATION

2. Draw diagrams of the pictures given below and also, write their functions.

ART INTEGRATION

a.



Ribcage

Function:

.....

.....

b.

Backbone

Function:

.....

.....

c.

Knee

Function:

.....

.....

Self-Assessment

Now that you have completed the chapter, score each of the following tasks from 1 to 5 to indicate how well you can do them.

Score 5 = I can definitely do this.

Score 1 = I cannot do this yet.

I can...	My score
• identify, name, and draw the major bones of the human body.	
• describe the functions of the major bones of the human body.	
• locate the major joints of the human body and discuss their functions.	
• draw diagrams of the shoulder and knee joints and their location in the body.	
• give examples of other kinds of joints in the human body.	
• explain what a muscle is, its types, and functions.	

Worksheet

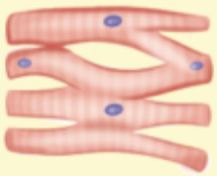
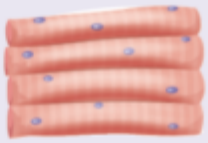
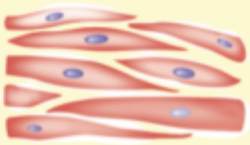
Observe the three types of muscles shown in the table given below. Firstly, give the definition and function of muscles. Then, complete the table as mentioned.

Definition of muscle:

.....

Function of muscle:

.....

S. no	Type of muscles	Drawing (to be drawn by the student)	Features
1.	 Name:		
2.	 Name:		
3.	 Name:		